

6. Examples of TI/TSC

Chern insulator is protected by U(1)_c charge conservation symmetry.

Fermion parity $\mathbb{Z}_2^f = \{1, P_f = (-1)^F\}$

$F = \sum_j c_j^\dagger c_j$ is the fermion number.

$$H = \sum c^\dagger c + (\Delta c c + \text{h.c.}) + V c^\dagger c^\dagger c c + \dots \text{ preserves } \mathbb{Z}_2^f.$$

6.1. 1+1D Majorana chain (by Kitaev 2000)

- Majorana fermion = real fermion

complex fermion $c, c^\dagger$:

$$\begin{cases} (c^\dagger)^2 = c^2 = 0 \\ c c^\dagger + c^\dagger c = 1 \end{cases}$$

Split c and $c^\dagger$ into "real" part and "imaginary" part:

$$\begin{cases} c = \frac{1}{2}(\gamma_1 + i\gamma_2) \\ c^\dagger = \frac{1}{2}(\gamma_1 - i\gamma_2) \end{cases} \quad \begin{cases} \gamma_1 = c + c^\dagger \\ \gamma_2 = \frac{1}{i}(c - c^\dagger) \end{cases}$$

Satisfying:

$$\begin{cases} \gamma_1^\dagger = \gamma_1, \gamma_2^\dagger = \gamma_2 \\ \gamma_1^2 = (c + c^\dagger)^2 = c c^\dagger + c^\dagger c = 1 \\ \gamma_2^2 = \frac{1}{i^2} (c - c^\dagger)^2 = c c^\dagger + c^\dagger c = 1 \\ \gamma_1 \gamma_2 + \gamma_2 \gamma_1 = \frac{1}{i} (c + c^\dagger)(c - c^\dagger) + \frac{1}{i} (c - c^\dagger)(c + c^\dagger) = 0 \end{cases}$$

$$\begin{cases} H = \epsilon c^\dagger c \Rightarrow \begin{cases} E = \epsilon, |E\rangle = |n=1\rangle, n = c^\dagger c = 1 \\ E = 0, |E\rangle = |n=0\rangle, n = c^\dagger c = 0 \end{cases} \\ n = c^\dagger c = \frac{1}{4}(\gamma_1 - i\gamma_2)(\gamma_1 + i\gamma_2) = \frac{1}{4}(2 + 2i\gamma_1\gamma_2) = \frac{1}{2}(1 + i\gamma_1\gamma_2) \\ P_f = (-1)^n = 1 - 2n = -i\gamma_1\gamma_2 = \pm 1 \\ H = \epsilon n = \epsilon \cdot \frac{1 - P_f}{2} = \frac{\epsilon}{2}(1 + i\gamma_1\gamma_2) \end{cases}$$

2 Majorana fermions $\gamma_1, \gamma_2 \longleftrightarrow$ 1 complex fermion mode c

$$\begin{array}{lll} \begin{array}{c} \xrightarrow{\gamma_1} \gamma_2 \end{array} \quad \begin{array}{c} (-i\gamma_1\gamma_2 = 1) \\ P_f = -i\gamma_1\gamma_2 = 1 \end{array} & n = c^\dagger c = 0 \\ \begin{array}{c} \xleftarrow{\gamma_2} \gamma_1 \end{array} \quad \begin{array}{c} (-i\gamma_2\gamma_1 = 1) \\ P_f = -i\gamma_1\gamma_2 = -1 \end{array} & n = c^\dagger c = 1 \end{array}$$

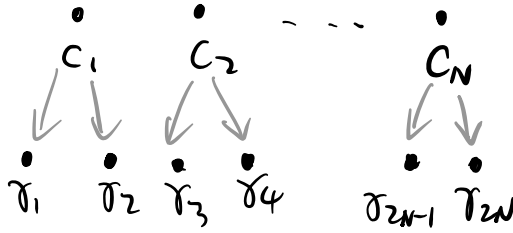
change pairing direction $\Leftrightarrow$ change fermion parity $P_f = \pm 1$

- Majorana chain.

Consider N complex fermions $\Rightarrow 2N$ Majorana fermions.

$$c_j = \frac{1}{2}(\gamma_{2j-1} + i\gamma_{2j})$$

$$(j=1, 2, \dots, N)$$



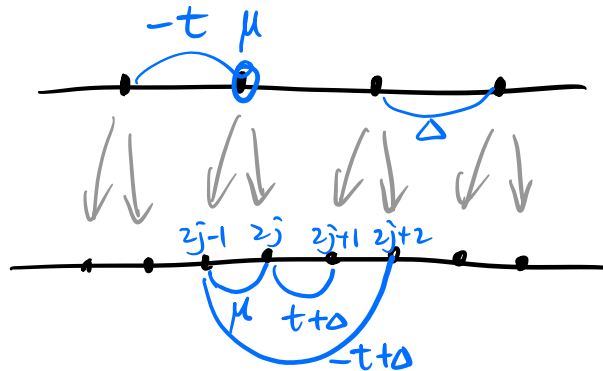
- (1) p-wave superconductor chain.

$$H = \sum_j \left[-t(c_j^\dagger c_{j+1} + \text{h.c.}) - \mu c_j^\dagger c_j + \Delta(c_j c_{j+1} + \text{h.c.}) \right]$$

periodic boundary $\rightarrow -t(c_N^\dagger c_1 + \text{h.c.})$ $t, \mu, \Delta \in \mathbb{R}$.

$$= \sum_j \left[-t \frac{1}{4} (\gamma_{2j-1} - i\gamma_{2j})(\gamma_{2j+1} + i\gamma_{2j+2}) + \text{h.c.} + \dots \right]$$

$$= \frac{i}{2} \sum_j \left[-\mu \gamma_{2j-1} \gamma_{2j} + (t+\Delta) \gamma_{2j} \gamma_{2j+1} + (-t+\Delta) \gamma_{2j-1} \gamma_{2j+2} \right]$$

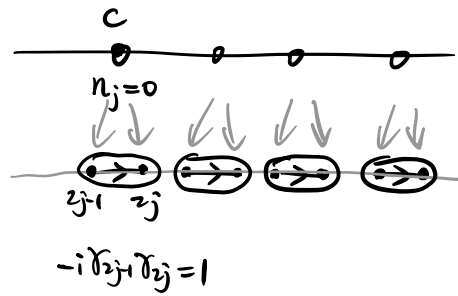


$$= \begin{cases} (\mu=-2, t=\Delta=0) = \sum_j i \gamma_{2j-1} \gamma_{2j} = - \sum_j (P_f)_j \\ (\mu=0, t=\Delta=1) = \sum_j i \gamma_{2j} \gamma_{2j+1} \end{cases}$$

- (2) Trivial chain ($\mu=-2, t=\Delta=0$)

$$H = - \sum_j (P_f)_j$$

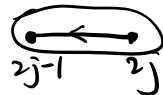
$$GS : n_j = 0 \Leftrightarrow (p_F)_j = +1 \\ \Leftrightarrow -i \gamma_{2j-1} \gamma_{2j} = 1$$



$$|GS\rangle = \bigotimes_j |n_j=0\rangle$$

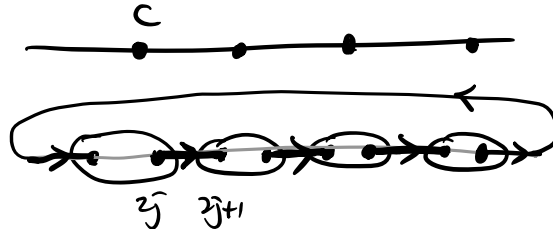
$$|GS\rangle = \bigotimes_j |-i \gamma_{2j-1} \gamma_{2j} = 1\rangle$$

$$ES : n_j = 1 \text{ for some } j.$$



(3) Nontrivial chain ($\mu=0, t=\delta=1$)

$$H = -\sum_j (c_j^\dagger c_{j+1} + \text{h.c.}) + \sum_j (c_j c_{j+1} + \text{h.c.}) \\ = \sum_j i \gamma_{2j} \gamma_{2j+1}$$



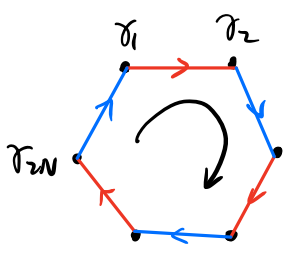
$$|GS\rangle = \bigotimes_j |-i \gamma_{2j} \gamma_{2j+1} = 1\rangle$$

$$\text{excited state: } \begin{array}{c} \leftarrow \\ 2j \quad 2j+1 \end{array} \quad -i \gamma_{2j} \gamma_{2j+1} = -1.$$

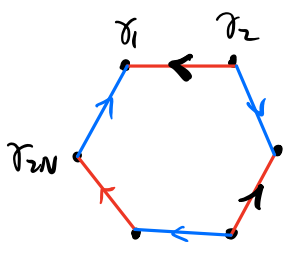
Fermion parity of $|GS\rangle$.

$$\begin{aligned} P_F |GS\rangle &= \prod_j (-i \gamma_{2j-1} \gamma_{2j}) |GS\rangle \\ &= (-i)^N (\gamma_1 \gamma_2) (\gamma_3 \gamma_4) \dots (\gamma_{2N-1} \gamma_{2N}) |GS\rangle \\ &= (-i)^N \gamma_1 (\gamma_2 \gamma_3) (\gamma_4 \gamma_5) \dots (\gamma_{2N-2} \gamma_{2N-1}) \gamma_{2N} |GS\rangle \\ &= (-i)^N (\gamma_2 \gamma_3) \dots (\gamma_{2N-2} \gamma_{2N-1}) (\gamma_1 \gamma_{2N}) |GS\rangle \\ &= -i \gamma_1 \gamma_{2N} |GS\rangle \\ &= -(-i \gamma_{2N} \gamma_1) |GS\rangle \\ &= -|GS\rangle \end{aligned}$$

$P_f = -1$ acting on $|\psi\rangle$
 $\Leftrightarrow |\psi\rangle$ has odd number of complex fermions c_j .



$$\begin{cases} |\psi_1\rangle: -i\sigma_1\sigma_2 = -i\sigma_3\sigma_4 = \dots = -i\sigma_{2N-1}\sigma_{2N} = 1 \\ |\psi_2\rangle: -i\sigma_2\sigma_3 = -i\sigma_4\sigma_5 = \dots = -i\sigma_{2N}\sigma_1 = 1 \\ |\psi_i\rangle \text{ and } |\psi_j\rangle \text{ have different fermion parity.} \end{cases}$$

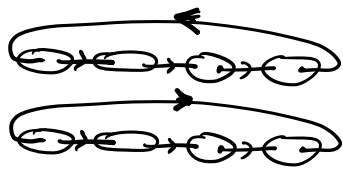
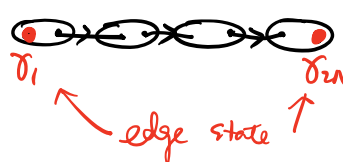


$|\psi_1\rangle$ and $|\psi_2\rangle$ have $\begin{cases} \text{same} \\ \text{different} \end{cases}$ fermion parity

$\Leftrightarrow$ number of counter clockwise arrow N_{ac} is $\begin{cases} \text{odd} \\ \text{even} \end{cases}$

$\Leftrightarrow$ loop is $\begin{cases} \text{Kastelyn} \\ \text{non-Kastelyn} \end{cases}$ oriented

$\begin{cases} \text{periodic boundary condition} \\ \text{antiperiodic} \quad \dots \\ \text{open} \quad \dots \end{cases}$

$P_f = -1$

$P_f = 1$

σ_1, σ_{2N} are unpaired Majorana fermion.
 $GSD = 2$

- p-wave SC in momentum space.

$$H = \sum_j \left[-t(c_j^\dagger c_{j+1} + \text{h.c.}) - \mu c_j^\dagger c_j + \Delta(c_j c_{j+1} + \text{h.c.}) \right]$$

$$\begin{cases} c_j = \frac{1}{\sqrt{N}} \sum_k e^{ikj} c_k \\ c_j^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{-ikj} c_k^\dagger \end{cases}$$

$$= \sum_k \left[-t(e^{ik} c_k^\dagger c_k + \text{h.c.}) - \mu c_k^\dagger c_k + \Delta(e^{ik} c_k c_k + \text{h.c.}) \right]$$

$$\sum_k \Delta e^{ik} c_k c_k$$

$$= \frac{1}{2} \sum_k \Delta e^{ik} c_k c_k + \frac{1}{2} \sum_k \Delta e^{-ik} c_k c_k$$

$$= \frac{1}{2} \sum_k (\Delta e^{ik} c_{-k} c_k - \Delta e^{-ik} c_{-k} c_k) \\ = \sum_k i \Delta \sin k c_{-k} c_k$$

$$= \sum_k (-2t \cos k - \mu) c_k^\dagger c_k + \Delta (i \sin k c_{-k} c_k + \text{h.c.}) \\ = \sum_k \frac{1}{2} (c_k^\dagger, c_{-k}) \underbrace{\begin{pmatrix} -2t \cos k - \mu & -i \Delta \sin k \\ 2i \Delta \sin k & 2t \cos k + \mu \end{pmatrix}}_{\mathcal{H}_k} \underbrace{\begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}}_{\text{Nambu basis}}$$

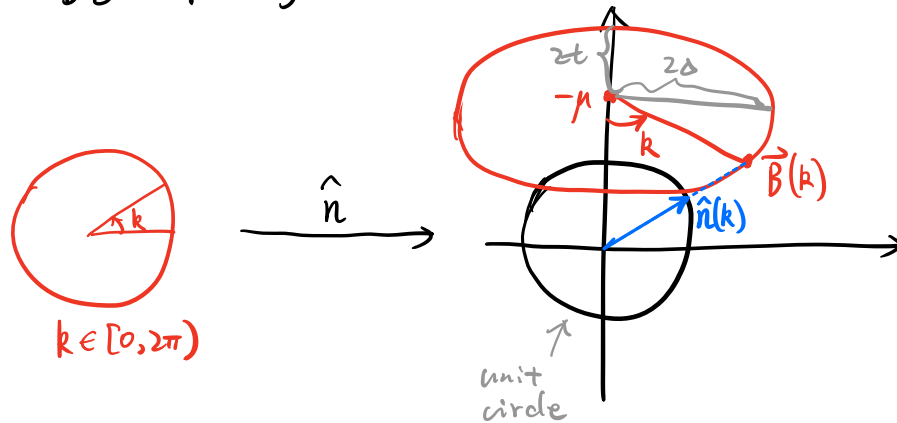
$$\mathcal{H}_k = 2\Delta \sin k \sigma_y - (2t \cos k + \mu) \sigma_z$$

$$= \vec{B}(k) \cdot \vec{\sigma} = E(k) \hat{n}(k) \cdot \vec{\sigma}$$

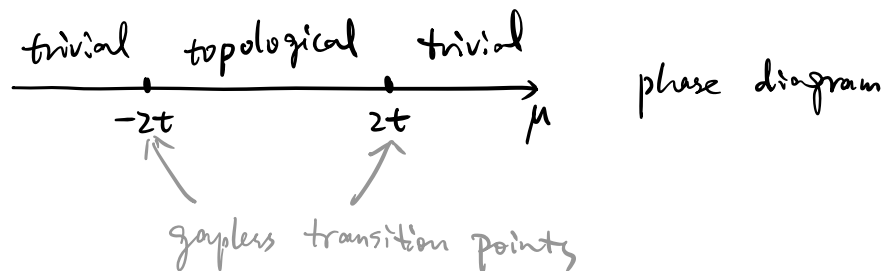
$$\vec{B}(k) := (2\Delta \sin k, -2t \cos k - \mu)$$

$$\hat{n}(k) := \frac{\vec{B}(k)}{|\vec{B}(k)|}$$

$$\hat{n} : BZ = T' = S' \longrightarrow S^1$$



$$\text{mapping degree} = \text{winding number} = \begin{cases} 1, & \text{if } -\mu - 2t < 0 < -\mu + 2t \\ 0, & \text{otherwise} \end{cases}$$



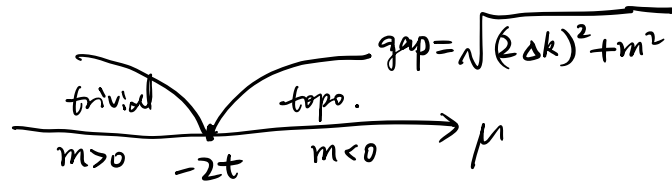
- Continuum model, field theory.

Consider $\mu = -2t - m$ with m small.

$$\mathcal{H}_k = 2\Delta \sin k \sigma_y + (-2t \cos k + 2t + m) \sigma_z$$

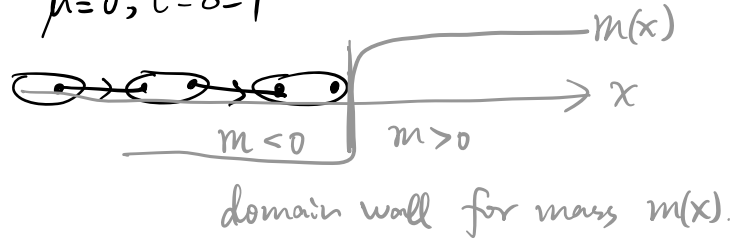
$$\xrightarrow{k \rightarrow 0} 2\Delta k \sigma_y + m \sigma_z$$

1+1 D massive Dirac Hamiltonian.



• Majorana edge mode

Consider $\mu=0, t=0=1$



For $\mu = -2t - m \approx -2t$:

$$\mathcal{H}_k = 2\Delta \cdot k \sigma_y + m \sigma_z$$

↓

$$\mathcal{H} = 2\Delta \sigma_y i\partial_x + m(x) \sigma_z$$

Try to solve zero energy state near $x=0$:

$$\mathcal{H} \psi(x) = 0$$

$$\Leftrightarrow 2\Delta \sigma_y i\partial_x \psi(x) + m(x) \sigma_z \psi(x) = 0 \quad v \sim \frac{\Delta}{m}$$

$$\Leftrightarrow \partial_x \psi(x) = -i \frac{1}{v} m(x) \sigma_x \psi(x)$$

$$\text{choose } \psi(x) = \varphi(x) \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$$

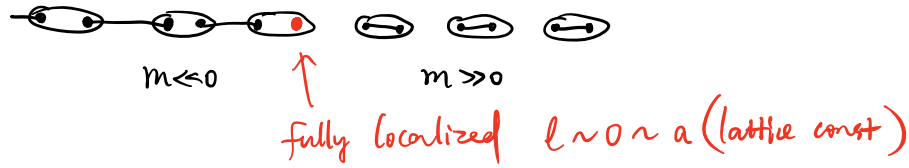
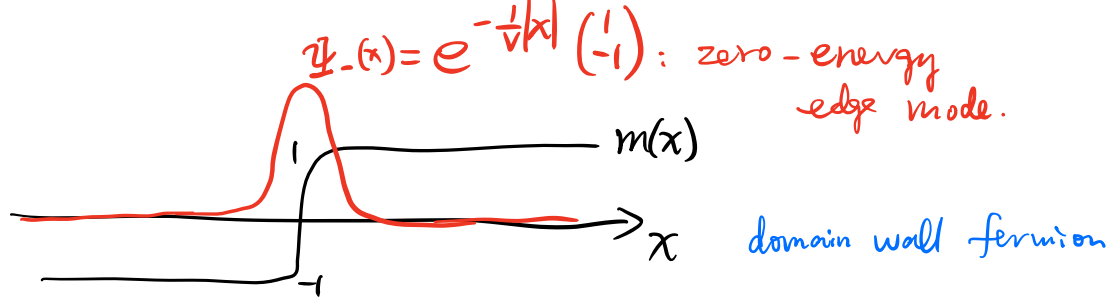
$$\Leftrightarrow \partial_x \varphi(x) = \pm \frac{1}{v} m(x) \varphi(x)$$

$$\Leftrightarrow \varphi(x) = e^{\pm \frac{1}{v} \int_0^x m(x') dx'}$$

$$\Leftrightarrow \psi(x) = e^{\pm \frac{1}{v} \int_0^x m(x') dx'} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$$

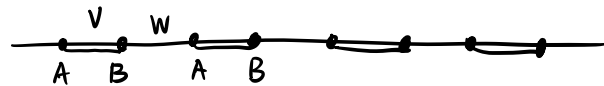
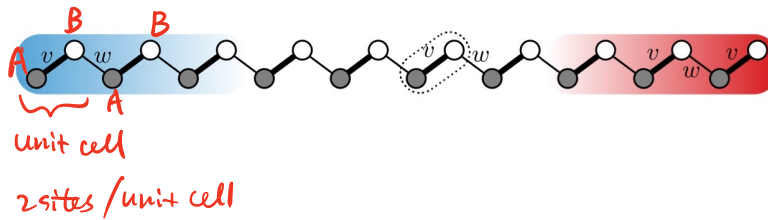
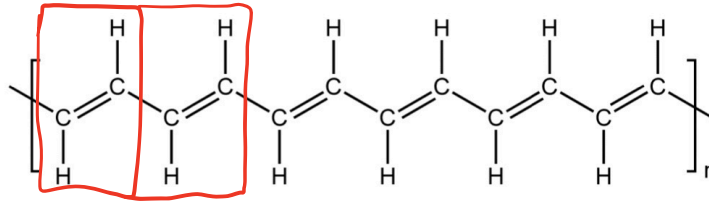
$$\begin{cases} \textcircled{1} \psi_+(x) = e^{+\frac{1}{v} \int_0^x m(x') dx'} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \xrightarrow{x \rightarrow \infty} e^{\frac{1}{v} x} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad x \\ \textcircled{2} \psi_-(x) = e^{-\frac{1}{v} \int_0^x m(x') dx'} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \xrightarrow{x \rightarrow -\infty} e^{-\frac{1}{v} x} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \checkmark \end{cases}$$

$v \sim \frac{\Delta}{m}$



6.2. 1D SSH model

SSH = Su-Schrieffer-Heeger



$$H = \sum_i \left(v c_{iA}^\dagger c_{iB} + w c_{iB}^\dagger c_{i+1A} + \text{h.c.} \right)$$

$$= \sum_k (c_{kA}^\dagger, c_{kB}^\dagger) \underbrace{\begin{pmatrix} 0 & v + w e^{-ik} \\ v + w e^{ik} & 0 \end{pmatrix}}_{\mathcal{H}_k} \begin{pmatrix} c_{kA} \\ c_{kB} \end{pmatrix}$$

$$\mathcal{H}_k = (v + w \cos k) \sigma_x + w \sin k \sigma_y$$

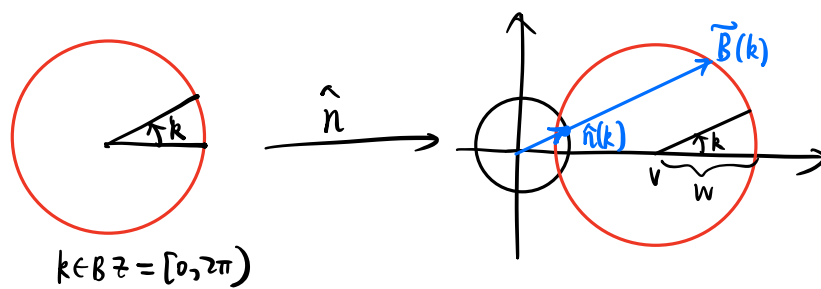
$$= \vec{B}(k) \cdot \vec{\sigma}$$

$$= E(k) \hat{n}(k) \cdot \vec{\sigma}$$

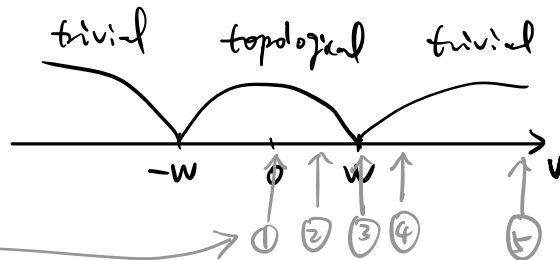
$$\vec{B}(k) = (v + w \cos k, w \sin k)$$

$$E(k) := |\vec{B}(k)| = \sqrt{v^2 + w^2 + 2vw \cos k}$$

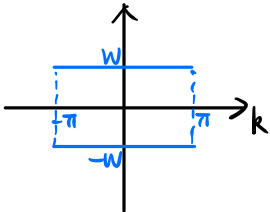
$$\hat{n}(k) := \vec{B}(k) / |\vec{B}(k)|$$



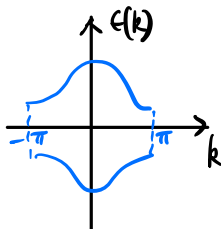
mapping degree = winding number = $\begin{cases} 1, & \text{if } V - W < 0 < V + W \\ 0, & \text{otherwise} \end{cases}$



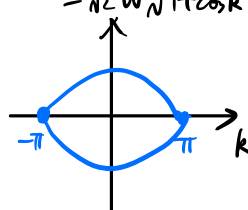
① $V=0$,
 $\epsilon(k) = \pm E(k) = \pm W$



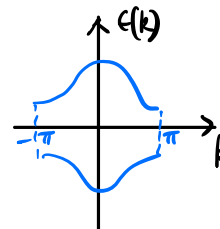
② $0 < V < W$



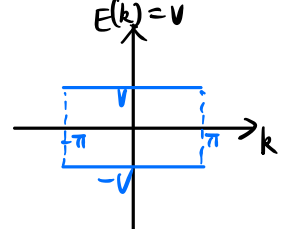
③ $V=W$
 $E(k) = \sqrt{2W^2 + 2W^2 \cos k}$
 $= \sqrt{2} W \sqrt{1 + \cos k}$



④ $0 < V < W$



⑤ $V > W$



bulk: topological

topological

trivial

trivial

edge: fully localized
 $l \sim 0 \sim a$

localized
 $l \sim \frac{1}{m}$

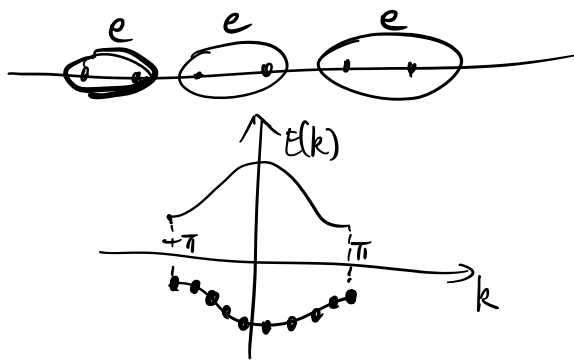
no edge states.

Similar to 1D Majorana chain (p-wave SC)!

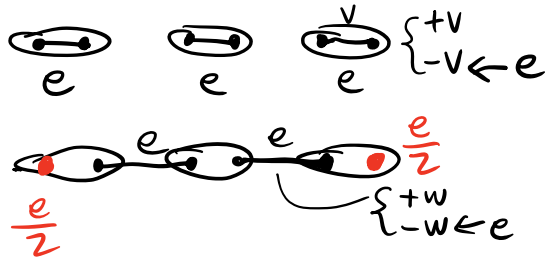
Differences:

Symm.: $\begin{cases} \text{Majorana chain has } \mathbb{Z}_2^f \text{ symmetry.} \\ \text{SSH has } U(1)_f \supset \mathbb{Z}_2^f \text{ symmetry.} \end{cases}$

edge: $\begin{cases} \text{unpaired Majorana fermion} \sim \text{half complex fermion} \sim \text{half } \mathbb{Z}_2^f \text{ charge} \\ \text{SSH edge state} \sim \text{half } U(1)_f \text{ charge.} \end{cases}$



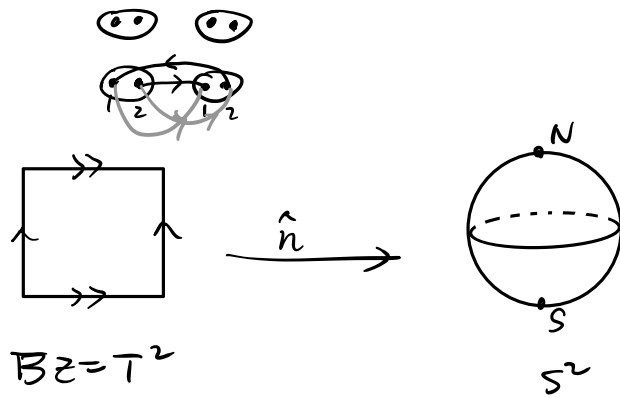
N unit cell $\left. \begin{array}{l} \\ N \text{ electrons} \end{array} \right\} \Rightarrow \text{insulator}$



6.3. Another model for 2D Chern insulator.

$$\mathcal{H}_{\vec{k}} = \sin k_x \sigma_x + \sin k_y \sigma_y + (u + \cos k_x + \cos k_y) \sigma_z$$

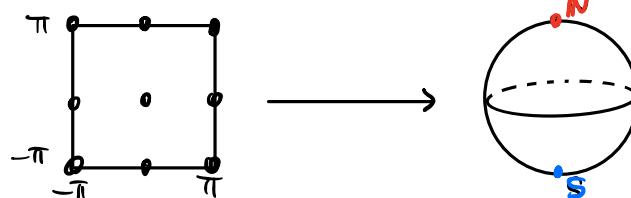
$$= \vec{B}(\vec{k}) \cdot \vec{\sigma}$$

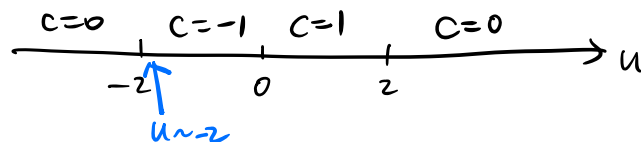
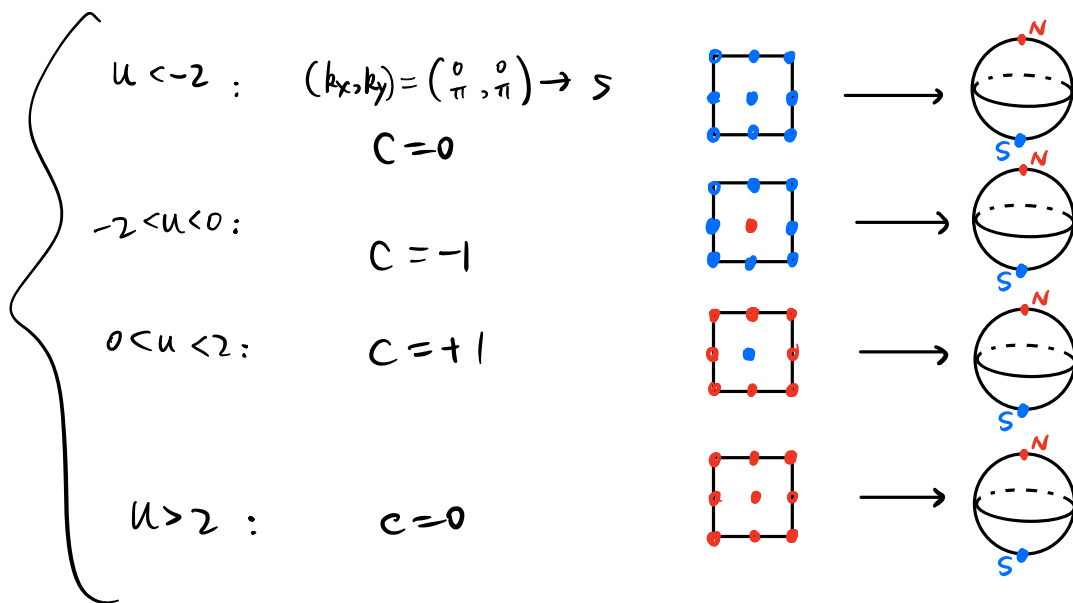


$$C = \text{mapping degree} = \sum_{\vec{k}_0 \in \hat{n}^{-1}(N_S)} \text{sgn det } D(\hat{n}(\vec{k}_0))$$

$$\hat{n}(\vec{k}_0) = \begin{cases} N \\ S \end{cases} = \begin{cases} (0, 0, 1) \\ (0, 0, -1) \end{cases} \Leftrightarrow \begin{cases} \sin k_x = 0 \Leftrightarrow k_x = 0, \pi \\ \sin k_y = 0 \Leftrightarrow k_y = 0, \pi \\ u + \cos k_x + \cos k_y > 0 \\ u + \cos k_x + \cos k_y < 0 \end{cases}$$

$[-2, 2]$



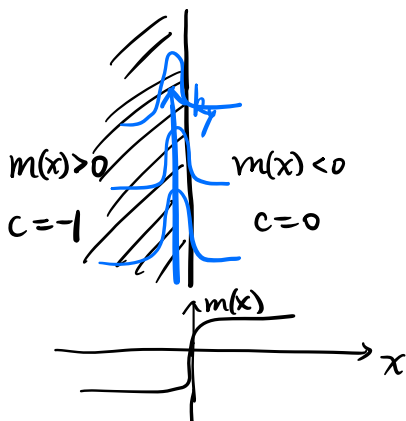


Edge state of the Chern insulator :

At $u \sim -2$, $\vec{k} \sim (0, 0)$

$$\begin{aligned}
 \mathcal{H}_{\vec{k}} &= \sin k_x \sigma_x + \sin k_y \sigma_y + (u + \cos k_x + \cos k_y) \sigma_z \\
 &\sim k_x \sigma_x + k_y \sigma_y + \underbrace{(u + 1 + 1)}_{m \sim 0} \sigma_z \\
 &= k_x \sigma_x + k_y \sigma_y + m \sigma_z \quad \rightarrow \text{2+1D Dirac Hamiltonian.}
 \end{aligned}$$

$$\begin{cases} m < 0 : C = 0 \text{ trivial} \\ m > 0 : C = -1 \text{ topological} \end{cases}$$



$$\mathcal{H}(x, k_y) = -i \sigma_x \partial_x + k_y \sigma_y + m(x) \sigma_z$$

Consider first $k_y = 0$ zero energy state:

$$\mathcal{H}(x, k_y=0) = -i \sigma_x \partial_x + m(x) \sigma_z \quad (1D \text{ Majorana chain edge problem})$$

$$\Rightarrow -i \sigma_x \partial_x \psi_{k_y=0}(x) = -m(x) \sigma_x \psi_{k_y=0}(x)$$

$$\Rightarrow \partial_x \psi_{k_y=0}(x) = -m(x) \sigma_y \psi_{k_y=0}(x)$$

$$\Rightarrow \psi_{k_y=0}(x) = e^{-|x|} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

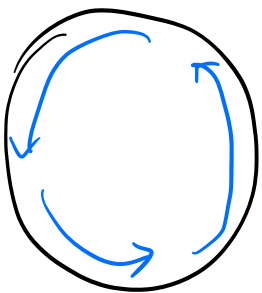
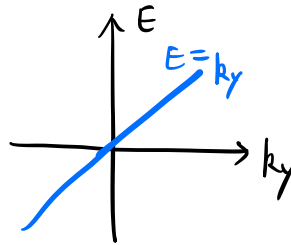
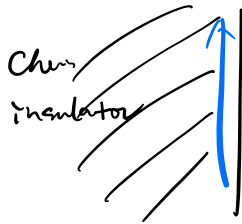
Consider k_y :

$$\psi_{k_y}(x) = e^{-|x|} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

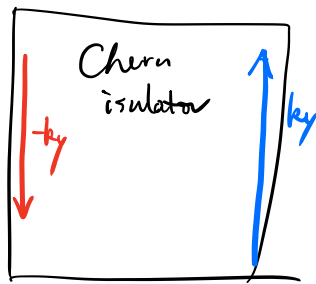
$$\Leftrightarrow \psi(x, y) = \underbrace{e^{-|x|}}_x \underbrace{e^{ik_y y}}_y \begin{pmatrix} 1 \\ i \end{pmatrix} \quad \text{decoupled}$$

$$\Rightarrow \partial_{k_y}(x) \psi(x, y) = k_y \sigma_y \psi(x, y) = k_y \psi(x, y)$$

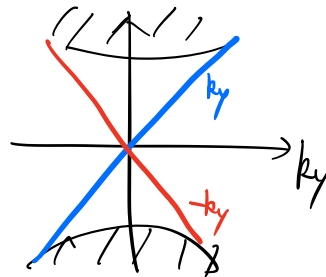
$$\Rightarrow \psi(x, y) \text{ is eigenvector of } \mathcal{H} \text{ with } E_R = k_y$$



1+1D chiral fermion



coupling $\sim e^{-Lx/3}$



6.4. 2D quantum spin Hall effect.

- Time reversal symmetry.

$$\begin{cases} \mathcal{T} \hat{H} \mathcal{T}^{-1} = \hat{H} & \hat{H} \text{ is invariant under } \mathcal{T}. \\ \mathcal{T} i \mathcal{T}^{-1} = -i & \mathcal{T} \text{ is anti-unitary.} \end{cases}$$

Choose a basis, $\mathcal{T} = T \cdot K$

↑
Unitary matrix complex conjugation operator

$$\mathcal{T}^2 = e^{i\phi} I$$

$$\Rightarrow \mathcal{T} \underbrace{K T K} = e^{i\phi}$$

$$\Rightarrow \mathcal{T} \mathcal{T}^* = e^{i\phi}$$

$$\Rightarrow \mathcal{T}^* = e^{i\phi} \mathcal{T}^\dagger = e^{i\phi} (\mathcal{T}^*)^\dagger$$

$$\Rightarrow \mathcal{T}^* = e^{i\phi} (\mathcal{T}^*)^\dagger = e^{i\phi} (e^{i\phi} (\mathcal{T}^*)^\dagger)^\dagger = e^{2i\phi} \mathcal{T}^*$$

$$\Rightarrow e^{2i\phi} = 1$$

$$\Rightarrow \mathcal{T}^2 = \mathcal{T} \mathcal{T}^* = \pm 1$$

$$\mathcal{T}^* = \pm \mathcal{T}^\dagger = (\pm \mathcal{T}^\dagger)^* \Rightarrow \mathcal{T}^\dagger = \pm \mathcal{T}$$

$$\begin{cases} \mathcal{T}^2 = +1 \Leftrightarrow \mathcal{T}^\dagger = \mathcal{T}, \mathcal{T} \text{ is symmetric} \\ \mathcal{T}^2 = -1 \Leftrightarrow \mathcal{T}^\dagger = -\mathcal{T}, \mathcal{T} \text{ is skew-symmetric.} \end{cases}$$

Example. For spin- $\frac{1}{2}$ system, $\mathcal{T}: \vec{S} \rightarrow -\vec{S}$, $\vec{S} = \frac{1}{2} \vec{\sigma}$

$$\mathcal{T} = T \cdot K$$

$$\mathcal{T} \frac{\vec{\sigma}}{2} \mathcal{T}^{-1} = -\frac{\vec{\sigma}}{2}$$

$$\Rightarrow \mathcal{T} = i\sigma_y K$$

$$\Rightarrow \mathcal{T}^2 = -1$$

For spin- S system, $\mathcal{T}: \vec{S} \rightarrow -\vec{S}$, $\vec{S} = \sum_{j=1}^{2S} \frac{1}{2} \vec{\sigma}_j$

$$\Rightarrow \mathcal{T}^2 = (-1)^{2S} = \begin{cases} 1, & S \text{ integer} \\ -1, & S \text{ half-odd-integer} \end{cases}$$

Kramers doublet :

For $\mathcal{T}$ invariant system with $\mathcal{T}^2 = -1$, $|a\rangle$ and $|b\rangle = \mathcal{T}|a\rangle$ are two different states with the same energy.

$$\left(\begin{array}{l} \text{If } \mathcal{T}|a\rangle = e^{i\phi}|a\rangle, \text{ then} \\ \mathcal{T}^2|a\rangle = \mathcal{T}(\mathcal{T}|a\rangle) = \mathcal{T} e^{i\phi}|a\rangle = e^{-i\phi} \mathcal{T}|a\rangle = e^{-i\phi} e^{i\phi}|a\rangle = |a\rangle \end{array} \right)$$

$$\Rightarrow \gamma^2 = +1$$

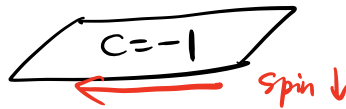
• 2D quantum Spin Hall effect.

QHE / Chern insulator breaks $\mathcal{T}$:

$$\vec{B} \rightarrow -\vec{B}$$

$$\text{chiral current: } \vec{j} \rightarrow -\vec{j}$$

For spin- $\frac{1}{2}$ system,



$$\mathcal{H}_k = \sin k_x \sigma_x + \sin k_y \sigma_y + (u + \cos k_x + \cos k_y) \sigma_z$$



$$\mathcal{H}_k = \sin k_x \underbrace{S_z \otimes \sigma_x}_{\substack{\uparrow \\ \text{spin}}} + S_0 \otimes \underbrace{[\sin k_y \sigma_y + (u + \cos k_x + \cos k_y) \sigma_z]}_{\substack{\uparrow \\ \text{band}}}$$

BHZ
||

Bernevig-Hughes-Zhang

$\{S_z \otimes \sigma_x, S_0 \otimes \sigma_y, S_0 \otimes \sigma_z\}$ anticommute with each other,

$$= \begin{pmatrix} \sin k_x \sigma_x + \sin k_y \sigma_y + (u + \cos k_x + \cos k_y) \sigma_z & 0 \\ 0 & -\sin k_x \sigma_x + \sin k_y \sigma_y + (u + \cos k_x + \cos k_y) \sigma_z \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} C=1 \\ C=-1 \end{pmatrix}$$

